

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.

```
import pandas as pd
```

```
# Create a small example dataset
```

```
data_dict = {  
    "Outlook": ["Sunny", "Sunny", "Overcast", "Rain"],  
    "Temperature": ["Hot", "Hot", "Hot", "Mild"],  
    "Humidity": ["High", "High", "High", "High"],  
    "Wind": ["Weak", "Strong", "Weak", "Weak"],  
    "PlayTennis": ["No", "No", "Yes", "Yes"]  
}
```

```
data = pd.DataFrame(data_dict)
```

```
# Save it to CSV (optional)
```

```
data.to_csv("training_data.csv", index=False)
```

```
# Step 1: Load the CSV file
```

```
data = pd.read_csv("training_data.csv")
```

```
# Step 2: Separate features and target
```

```
features = data.columns[:-1] # All columns except last
```

```
target = data.columns[-1]    # Last column is target
```

```
# Step 3: Initialize the most specific hypothesis
```

```
# Using '0' to indicate most specific (no generalization yet)
```

```
hypothesis = ['0'] * len(features)
```

```
# Step 4: Find-S algorithm
```

```
for index, row in data.iterrows():
```

```
if row[target] == 'Yes': # Only consider positive examples
    for i, feature in enumerate(features):
        if hypothesis[i] == '0': # If most specific, take the value
            hypothesis[i] = row[feature]
        elif hypothesis[i] != row[feature]: # Generalize if mismatch
            hypothesis[i] = '?'
```

Step 5: Output the final hypothesis

```
print("Most specific hypothesis consistent with the data:")
print(hypothesis)
```

output:

```
Most specific hypothesis consistent with the data:
['?', '?', 'High', 'Weak']
```